

Analyst / Technical Report —

CHALLENGE-NOTCHITUP

Likelihood scale (FIRST 5-tier): almost_certain > highly_likely > likely > unlikely > remote. Likelihood estimates the probability of the assessed activity; it is kept separate from analytic confidence.

Confidence (LCA): high / moderate / low — the analyst's confidence in the assessment given evidence quality and corroboration. Distinct from likelihood.

Findings

PAGE_EXECUTE_READWRITE injected region in explorer.exe (PID 1944)

- **Finding ID:** F-NOTCH-001 · **Severity:** high · **Likelihood:** unlikely · **Confidence:** high · **Risk score:** 8
- **MITRE ATT&CK:** T1055

malfind flagged a 64KB PAGE_EXECUTE_READWRITE VAD at 0x4320000 in explorer.exe (PID 1944) carrying executable indirect-jump shellcode bytes (41 ba 80 00 00 00 48 b8 ... 48 ff 20). Classic code-injection signature.

Evidence: windows.malfind: PID 1944 explorer.exe VAD 0x4320000
PAGE_EXECUTE_READWRITE 65536B (executable bytes 41 ba 80 00 00 00 48 b8 ...)

carved region payload_sha256

65196e1a65d8e4bfcf42f03b7db79cd07a2573f57c6aad40a97c37791726ca6f

source_evidence_sha256

80366d7ec64a5529c95c2f523f4281a5f11efbad33ecb19f73525470c1407b23

PAGE_EXECUTE_READWRITE zeroed RWX region in explorer.exe (PID 1944)

- **Finding ID:** F-NOTCH-002 · **Severity:** medium · **Likelihood:** unlikely · **Confidence:** moderate · **Risk score:** 6
- **MITRE ATT&CK:** T1055

malfind flagged a second 4KB PAGE_EXECUTE_READWRITE VAD at 0x3ce0000 in explorer.exe (PID 1944), zeroed - a second RWX VAD in the same injected host process.

Evidence: windows.malfind: PID 1944 explorer.exe VAD 0x3ce0000
PAGE_EXECUTE_READWRITE 4096B (zeroed)
carved region payload_sha256
3da12179ac97a8fdcfbdfc5318ba0da46974e1c1ae648dc37e861a926e8563ca
source_evidence_sha256
80366d7ec64a5529c95c2f523f4281a5f11efbad33ecb19f73525470c1407b23

PAGE_EXECUTE_READWRITE RWX region in chrome.exe (PID 2124)

- **Finding ID:** F-NOTCH-003 · **Severity:** medium · **Likelihood:** unlikely · **Confidence:** moderate · **Risk score:** 6
- **MITRE ATT&CK:** T1055

malfind flagged a 4KB PAGE_EXECUTE_READWRITE VAD at 0x4830000 in chrome.exe (PID 2124), zeroed.

Evidence: windows.malfind: PID 2124 chrome.exe VAD 0x4830000
PAGE_EXECUTE_READWRITE 4096B (zeroed)
carved region payload_sha256
3243bcc13c7c564288355046c5bc1123888837aeb8ada9ed5cf7800d9c462c1d
source_evidence_sha256
80366d7ec64a5529c95c2f523f4281a5f11efbad33ecb19f73525470c1407b23

Large PAGE_EXECUTE_READWRITE region in WmiPrvSE.exe (PID 2292)

- **Finding ID:** F-NOTCH-004 · **Severity:** high · **Likelihood:** unlikely · **Confidence:** high · **Risk score:** 8
- **MITRE ATT&CK:** T1047, T1055

malfind flagged a 512KB PAGE_EXECUTE_READWRITE VAD at 0x1bd0000 in WmiPrvSE.exe (PID 2292) - WMI provider host carrying a large RWX region.

Evidence: windows.malfind: PID 2292 WmiPrvSE.exe VAD 0x1bd0000
PAGE_EXECUTE_READWRITE 524288B (512KB)
carved region payload_sha256
75b4c5d8fc85525f97106ddf84197731942bb9acb2d992de50b8a4b7bcfad9a2
source_evidence_sha256
80366d7ec64a5529c95c2f523f4281a5f11efbad33ecb19f73525470c1407b23

Evidence image registered (chain-of-custody hash)

• **Finding ID:** F-NOTCH-005 · **Severity:** info · **Likelihood:** unlikely · **Confidence:** high · **Risk score:** 0

evidence_register computed the chain-of-custody SHA-256 for the 1.6GB raw memory image.

Evidence: evidence_register: /cases/Challenge_NotchItUp/Challenge.raw
sha256
80366d7ec64a5529c95c2f523f4281a5f11efbad33ecb19f73525470c1407b23
size 1610547200 bytes
source_evidence_sha256
80366d7ec64a5529c95c2f523f4281a5f11efbad33ecb19f73525470c1407b23

Indicators of Compromise

Value	Type
explorer.exe:PID1944:0x4320000:RWX	memory_re
WmiPrvSE.exe:PID2292:0x1bd0000:RWX	memory_re
explorer.exe	process
WmiPrvSE.exe	process
80366d7ec64a5529c95c2f523f4281a5f11efbad33ecb19f73525470c1407b23	sha256

(provenance graph omitted in PDF — see the IOC table above; renders in the HTML/GitLab view)

Timeline

(provenance graph omitted in PDF — see the IOC table above; renders in the HTML/GitLab view)

Timestamp	Host	Event	Phase	Description
2019-08-19T14:40:07Z	notch-it-up	TL-NOTCH-001	—	System boot - System (PID 4) and smss.exe (PID 264) started; Windows x64 kernel symbols matched by Volatility3 (53 processes)
2019-08-19T14:40:11Z	notch-it-up	TL-NOTCH-002	—	Core Windows service stack started: wininit.exe (384), services.exe (480), lsass.exe (496), svchost.exe (608), VBoxService.exe (668) - VirtualBox guest
2019-08-19T14:40:07Z	notch-it-up	TL-NOTCH-003	—	Code injection observed: explorer.exe (PID 1944) holds RWX VAD at 0x4320000 with 64KB executable shellcode (41 ba ... 48 b8 ...) and a zeroed RWX VAD at 0x3ce0000
2019-08-19T14:40:07Z	notch-it-up	TL-NOTCH-004	—	WmiPrvSE.exe (PID 2292) carries a 512KB RWX VAD at 0x1bd0000; chrome.exe (PID 2124) holds a 4KB RWX VAD at 0x4830000
2019-08-19T14:40:11Z	notch-it-up	TL-NOTCH-005	—	Active browser network sessions in memory: firefox.exe (PID 2080) ESTABLISHED to Google ranges (172.217.160.131:80, 172.217.194.189:443) from guest 10.0.2.15; chrome.exe (PID 2124) UDP 5353 mDNS